

DR SHETYE ACADEMIC PROGRAMME

Maths Level 2 Test — August 3rd, 2026

CAT / IIM Level · PYQ-Style · 30 Questions · 30 Marks · 45 Minutes

COMPLETE SOLUTIONS & ANSWER KEY

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	D	3	A	4	B	5	D
6	A	7	D	8	A	9	B	10	D
11	A	12	B	13	D	14	A	15	D
16	B	17	A	18	B	19	A	20	B
21	B	22	C	23	D	24	B	25	C
26	C	27	C	28	C	29	B	30	D

■■ PROFIT / LOSS & COMMERCIAL

Q1. A trader buys for ₹1,400 and sells for ₹1,260. Loss%?

- A) 10%
- B) 9%
- C) 8%
- D) 12%

✓ **Answer: (A) 10%**

Solution: Loss = 140. Loss% = $(140/1400) \times 100 = 10\%$.

Q2. Sells at 10% loss. ₹150 more → 5% profit. Find CP.

- A) ₹800
- B) ₹900
- C) ₹1,100
- D) ₹1,000

✓ **Answer: (D) ₹1,000**

Solution: $0.15x = 150 \rightarrow CP = ₹1,000$. Verify: SP at loss = 900; at profit = 1050; diff = 150 ✓.

Q3. Cheats 20% buying and 20% selling. Profit%?

- A) 50%
- B) 44%
- C) 40%
- D) 56%

✓ **Answer: (A) 50%**

Solution: Profit% = $(120/80) \times 100 - 100 = 150 - 100 = 50\%$.

■■ COMPOUND INTEREST & SIMPLE INTEREST

Q4. Difference between CI and SI on ₹5,000 at 8% for 2 years.

- A) ₹28
- B) ₹32
- C) ₹36
- D) ₹40

✓ **Answer: (B) ₹32**

Solution: Diff = $P \times (R/100)^2 = 5000 \times 0.0064 = ₹32$.

Q5. CI on a sum at 10% for 2 years = ₹420. Find principal.

- A) ₹1,800
- B) ₹1,900
- C) ₹2,100

D) ■2,000

✓ Answer: (D) ■2,000

Solution: $420 = P \times 0.21 \rightarrow P = \text{■}2,000$.

Q6. CI exceeds SI by ■50 for 2 years at 10%. Find principal.

A) ■5,000

B) ■4,500

C) ■4,000

D) ■5,500

✓ Answer: (A) ■5,000

Solution: $50 = P \times (0.1)^2 \rightarrow P = \text{■}5,000$.

■■ RATIO, PROPORTION & PARTNERSHIP

Q7. A:B = 3:4. A gets ■50 more, ratio $\rightarrow$ 5:6. Find A's original share.

A) ■350

B) ■400

C) ■500

D) ■450

✓ Answer: (D) ■450

Solution: $(3x+50)/4x = 5/6 \rightarrow x=150$. $A = 3 \times 150 = \text{■}450$.

Q8. Bill ■1,950 split in ratio 1/2:1/3:1/4. Largest share?

A) ■900

B) ■800

C) ■750

D) ■1,000

✓ Answer: (A) ■900

Solution: Ratio = 6:4:3. Largest = $1950 \times 6/13 = \text{■}900$.

■■ ALLIGATION & MIXTURES

Q9. Milk ■16/L, water free. Mix sold at ■12/L with 50% profit. Milk:Water ratio?

A) 1:2

B) 1:1

C) 2:1

D) 3:1

✓ Answer: (B) 1:1

Solution: CP of mixture = $12/1.5 = \text{■}8$. Alligation: $(16-8):(8-0) = 1:1$.

Q10. 35L wine:water = 4:3. Remove 7L, add 7L water. New ratio?

A) 12:19

B) 14:19

C) 18:19

D) 16:19

✓ Answer: (D) 16:19

Solution: Wine = $20 - 4 = 16\text{L}$. Water = $15 - 3 + 7 = 19\text{L}$. Ratio = 16:19.

■■ SPEED, DISTANCE & TIME

Q11. 2/5 journey at 40 km/h, 3/5 at 60 km/h. Average speed?

A) 50 km/h

B) 48 km/h

C) 44 km/h

D) 52 km/h

✓ Answer: (A) 50 km/h

Solution: Time■ = $D/100$, Time■ = $D/100$. Total = $D/50$. Avg = 50 km/h.

Q12. Train crosses pole in 15s, platform 300m in 45s. Speed?

- A) 30 km/h
- B) 36 km/h
- C) 40 km/h
- D) 45 km/h

✓ **Answer: (B) 36 km/h**

Solution: $L=150\text{m}$. Speed= $10\text{ m/s}=36\text{ km/h}$.

■ ■ TIME & WORK

Q13. $A+B=20d$, $B+C=30d$, $C+A=24d$. All three together?

- A) 12 days
- B) 14 days
- C) 18 days
- D) 16 days

✓ **Answer: (D) 16 days**

Solution: $2(A+B+C)=1/20+1/30+1/24=1/8$. Together=16 days.

Q14. Tap fills in 6h. Leak empties in 8h. Time to fill?

- A) 24 hours
- B) 22 hours
- C) 20 hours
- D) 28 hours

✓ **Answer: (A) 24 hours**

Solution: Net = $1/6-1/8 = 1/24$. Fill time = 24 hours.

■ ■ PERCENTAGE

Q15. Number: +20% then -25%. Net change?

- A) 5% decrease
- B) 8% decrease
- C) 12% decrease
- D) 10% decrease

✓ **Answer: (D) 10% decrease**

Solution: $1.20 \times 0.75 = 0.90 \rightarrow 10\%$ decrease.

Q16. 30% of A = 0.25 of B. Find A:B.

- A) 4:5
- B) 5:6
- C) 6:7
- D) 7:8

✓ **Answer: (B) 5:6**

Solution: $0.30A=0.25B \rightarrow A/B=5/6$. A:B=5:6.

■ ■ NUMBER THEORY

Q17. Find HCF of 72 and 120.

- A) 24
- B) 20
- C) 18
- D) 36

✓ **Answer: (A) 24**

Solution: $72=2^3 \times 3^2$, $120=2^3 \times 3 \times 5$. HCF= $2^3 \times 3=24$.

Q18. Remainder when $2^{\text{■■■}}$ is divided by 3.

- A) 0
- B) 1

- C) 2
- D) 3

✓ Answer: (B) 1

Solution: Period 2: $2^2 \equiv 1 \pmod{3}$. 200 even $\rightarrow 2^{200} \equiv 1$. Remainder=1.

Q19. Sum of first 50 even natural numbers.

- A) 2550
- B) 2500
- C) 2400
- D) 2600

✓ Answer: (A) 2550

Solution: $2+4+\dots+100 = 2(1+2+\dots+50) = 50 \times 51 = 2550$.

■■ PERMUTATIONS & COMBINATIONS

Q20. 6 people, committee of 4. Two specific must NOT both be included. Ways?

- A) 7
- B) 9
- C) 11
- D) 13

✓ Answer: (B) 9

Solution: Total $C(6,4)=15$. Both included: $C(4,2)=6$. Valid $=15-6=9$.

Q21. 3-digit even numbers from {1,2,3,4,5} without repetition.

- A) 18
- B) 24
- C) 30
- D) 36

✓ Answer: (B) 24

Solution: Units digit (2 or 4): 2 choices. Hundreds: 4. Tens: 3. Total $=2 \times 4 \times 3 = 24$.

■■ PROBABILITY

Q22. $P(A)=0.6$, $P(B)=0.5$, $P(A \cap B)=0.3$. Find $P(A|B)$.

- A) 0.4
- B) 0.5
- C) 0.6
- D) 0.7

✓ Answer: (C) 0.6

Solution: $P(A|B) = P(A \cap B) / P(B) = 0.3 / 0.5 = 0.6$.

■■ ALGEBRA

Q23. $a+b+c=9$, $ab+bc+ca=26$. Find $a^2+b^2+c^2$.

- A) 25
- B) 27
- C) 31
- D) 29

✓ Answer: (D) 29

Solution: $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$. $81 = a^2+b^2+c^2+52$. Answer=29.

Q24. Simplify: $(x^2-5x+6) \div (x-2)$.

- A) $x+3$
- B) $x-3$
- C) $x+2$
- D) $x-2$

✓ Answer: (B) $x-3$

Solution: $(x-2)(x-3)/(x-2)=x-3$, for $x \neq 2$.

■ ■ GEOMETRY & MENSURATION

Q25. Rectangle 6×8 cm. Circumscribed circle area. ($\pi=22/7$)

- A) 68.75 cm²
- B) 75.5 cm²
- C) 78.5 cm²
- D) 82 cm²

✓ **Answer: (C) 78.5 cm²**

Solution: Diagonal = $\sqrt{(6^2+8^2)}=10$. Radius = 5. Area = $(22/7) \times 25 \approx 78.5$ cm².

Q26. Volume of hemisphere radius 7 cm. ($\pi=22/7$)

- A) 680.5 cm³
- B) 700 cm³
- C) 718.67 cm³
- D) 735 cm³

✓ **Answer: (C) 718.67 cm³**

Solution: $(2/3) \times (22/7) \times 343 = 2156/3 \approx 718.67$ cm³.

■ ■ SETS, AVERAGES & STATISTICS

Q27. Average of 15 numbers = 48. Remove one, average → 50. Number removed?

- A) 14
- B) 18
- C) 20
- D) 22

✓ **Answer: (C) 20**

Solution: $15 \times 48 - 14 \times 50 = 720 - 700 = 20$.

Q28. 80 students: 45 cricket, 30 football, 20 both. Neither?

- A) 15
- B) 20
- C) 25
- D) 30

✓ **Answer: (C) 25**

Solution: At least one = $45 + 30 - 20 = 55$. Neither = $80 - 55 = 25$.

■ ■ LOGARITHMS & SEQUENCES

Q29. $\log(x+3) + \log(x-3) = \log 16$. Find x.

- A) x = 4
- B) x = 5
- C) x = 6
- D) x = 7

✓ **Answer: (B) x = 5**

Solution: $x^2 - 9 = 16 \rightarrow x^2 = 25 \rightarrow x = 5$ (domain: $x > 3$).

Q30. AP: 3rd term = 7, 7th term = 19. Sum of first 10 terms?

- A) 130
- B) 135
- C) 140
- D) 145

✓ **Answer: (D) 145**

Solution: $d = 3$, $a = 1$. $S_{10} = 5(2 + 27) = 5 \times 29 = 145$.